

Tie Memories to E-souvenirs: Hybrid Tangible AR Souvenirs in the Museum

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ABSTRACT

Traditional physical souvenirs in museums have three major limits: monotonous interaction, lack of personalization, and disconnection to the exhibition. To conquer these problems and to make personalized souvenirs a part of the visiting experience, we create a hybrid tangible Augmented Reality(AR) souvenir that combines a physical firework launcher and AR models. An application called AR Firework is designed for customizing the hybrid souvenir as well as interactive learning in an exhibition in the wild. Multiple interaction methods including mobile user interface, hand gestures, and voice are adopted to create a multi-sensory product. As the first research to propose tangible AR souvenirs, we find that they establish a long-lasting connection between visitors and their personal visiting experiences. This paper promotes the understanding of personalization, socialization and tangible AR.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Mixed/augmented reality**;

KEYWORDS

Souvenir, Personalization, Museums, Augmented Reality

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1 INTRODUCTION

Nowadays, a growing number of museums have adopted digital media to enhance their exhibitions. The post-visit[3] experience where souvenirs play an essential role is as important as the one in the museums. However, most souvenirs are noninteractive and monolithic, thus failing to build a long-lasting relationship between the museum and the visitors[3].

It has been revealed that humans can access and retrieve information that is linked with related objects easily[5]. Attaching digital information to physical materials has been studied in some previous works. Some researchers have been trying to make personalized souvenirs during their visits to museums. For example, Petrelli et al proposed the data tangible souvenir, a postcard with data recorded throughout the visit[3]. Our work is inspired by these projects. Whereas the core differences with them can be concluded to (1)We adopt AR for creating more-complex, personalized, and multi-sensory souvenirs. It combines all the information mentioned in previous personalized souvenirs like text, photos and audio. (2)Various interaction methods including AR, hand gestures, and voice are adopted to create multi-sensory products. (3)Creating personalized experiences in museums need complex artistic installations in most cases. We propose an application for mobile to address this problem. Meanwhile, we solve the problem brought by mobile guidance that people will focus on the screen, therefore neglect the exhibits. By overlaying virtual information in the real environment, AR redirects attention to the exhibition.

In our work, we propose the hybrid tangible AR souvenir called AR Firework. We create a system to produce it in the Shanghai Science and Technology Museum. It allows children to interact with the exhibits in the process of designing AR fireworks. In this paper, we give a detailed description of the design and report on the deployment of the system in the wild.

2 BACKGROUND

While purchasing souvenirs after the visit is a common way for people to bear a more lively memory, studies show that making the souvenir an integral part of the visit[3] is a better way to augment the experience. Tangible interfaces have been applied in creating emotional souvenirs, like Music Bottles made by Ishii et al[2], which combines the tangible property of the glass bottle and electronic music sounds. This combination brings about multi-sensory sensations and augments users' perception of the object[4].

Among various technologies, tangible augmented reality(AR) combine the enhanced display possibilities of AR with the intuitive manipulation and interaction of physical objects[1]. Despite its great potential, there are few studies on tangible AR gifts/souvenirs.

3 DESIGN

3.1 AR Firework system

We create a system for producing hybrid AR souvenirs for Shanghai Science and Technology Museum. It contains an application, a QR code printer system and an online gallery. Children use their phones as the display devices for AR to explore the exhibition and interact with the exhibits. They will learn lots of physical and chemical knowledge in the process of making fireworks. If they want to take the fireworks models as a souvenir, they can buy a physical fireworks launcher and an auto-generated souvenir from the salesmen/saleswomen. The personalized AR models and users' voices will be uploaded after scanning the QR code sticker on the box.

3.2 The hybrid souvenir with DIY firework

The hybrid tangible AR souvenir includes both physical objects and digital information. The physical part retains the formal sense of owning a souvenir. The virtual part contains AR models and audio that serves as a reminder of the visit. The virtual part is made by the visitor during his/her visiting process. In the exhibition, there are hidden materials necessary for making a firework. The visitors collect all the materials by scanning the ores in the exhibition hall with their phones, and then design the firework on their own. We provide several types of fireworks for users to choose from, including peonies, comets, etc. Users can adjust the parameters like height, color, diameter, etc. Except for these common fireworks, we developed a picture-based fireworks generation algorithm. In the VFX particle system, an average of 20,000 particles are used to simulate each firework. The pixels of picture1.a are transformed into sparks at the moment of the firework explosion. The particle trajectory under the gravity field is simulated. Users can take photos and convert them into part of their artwork. In this way, memories are evoked when users open the souvenirs again. After making the fireworks, users can decide the places to set the fireworks, for example, around a crystal ball(as figure1.b), in a castle, etc. Besides the traditional experience of "seeing", visitors are encouraged to record their voices and bind them to the souvenir as depicted in figure1.c. The voices can be their feelings towards the visit or blessing if they decide to give the souvenirs to others. This voice messages will be played along with the firework show. The physical part is an exquisite box as shown in figure2. Then users need to stick a QR code onto the box and bind the whole AR model they created. People who want to watch the show can download the AR models and audio with the AR Firework application by scanning the QR code. This souvenir can also promote socialization. The visitors can send this personalized firework souvenir as well as recorded voice messages of blessing as a gift to others. They can upload these fireworks to the public "fireworks gallery" in the application.

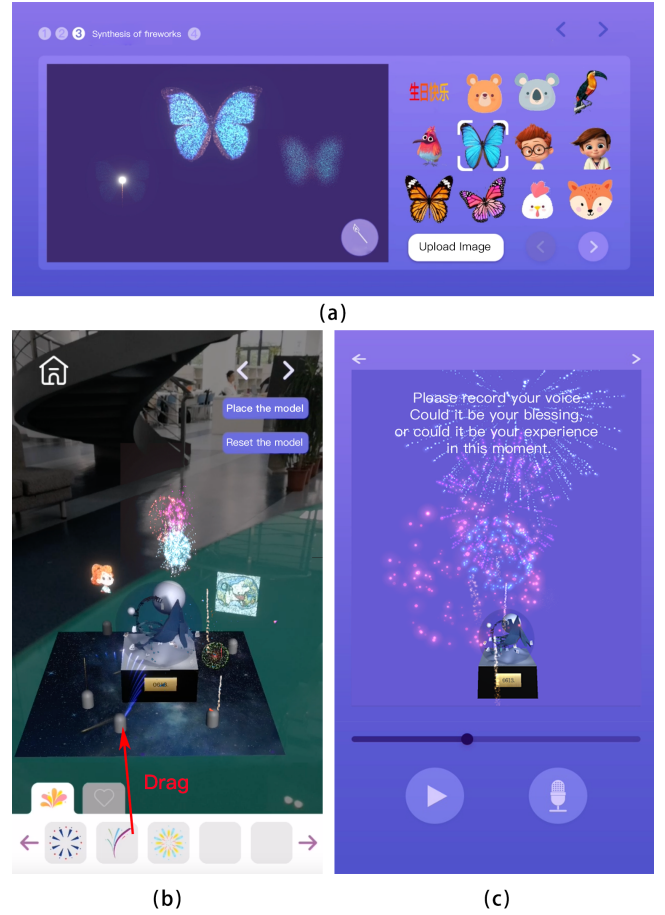


Figure 1: Three ways to increase the freedom to customize fireworks.

3.3 Multiple-sensory interaction and augmented reality

Every digital interaction is a sensory experience. The multi-sensory design makes the engagement more multifaceted and memorable. Vision and sound contribute to communication, while touch brings connections with the object. The visual experience is enhanced by AR technology. AR supported with ARFoundation SDK is used in various scenarios, including interaction with the exhibits, presentation of the 3D molecular models, simulation of the experiments, fireworks design process, etc. Instead of adopting the traditional AR mobile interface, which is dragging an object to a single plane, we made collision bodies for the models where the fireworks can be placed(cake, castle, etc.), so that customized fireworks can be dragged anywhere on the surface of the model and set off. In this way, the spatial sense of the model and the freedom of arrangement are improved. Sound promotes socialization. Users can record and bind their voices to the souvenir. It can be a blessing for a friend or their feeling about the exhibition. Physical firework launcher and hand gesture interaction enhance the touch sensory. Natural hand gesture detection supported by ManoMotion SDK[6] is used



Figure 2: The release gesture triggers the fireworks set off and the pinch gesture triggers the rotation of the model.

to trigger the launch of fireworks. ManoMotion supports five hand gestures including pick, pinch, click, grab and release. As shown in figure 2. We finally choose release as the trigger gesture, because setting off fireworks on users' hands looks like magic. In consideration of convenience and precision, we choose the pinch gesture to control the rotation. Controlling time span through thumb and forefinger contact is the most agile and natural gesture in the five gestures provided by ManoMotion. The model rotates clockwise when users pinch their fingers. It stops rotating when users release their fingers. Therefore, users can view AR models from multiple directions easily.

4 DISCUSSION

4.1 AR personalization

The customization of AR souvenirs mainly focuses on the models which have more possibility for creation and personalization. In the

proposed application, models are divided into given ones and DIY ones. The latter are generated by a particle system according to the custom parameters and photos. More research needs to be done on how to make artistic AR models that can carry richer information.

Moreover, we believe that AR souvenirs offer new possibilities for the existing market of gifts, souvenirs and packaging. Placing a marker and binding an AR model to the package provides a more vivid display of traditional gifts and souvenirs, as well as emphasizes the concept of socialization. With the development of object recognition technology, the binding of AR models and physical objects can be more natural.

5 CONCLUSION

As the first research to propose tangible AR hybrid souvenirs, we believe that it has great potential in future packaging, souvenirs, gifts, and other markets. In summary, The major two contributions are:

- We propose hybrid tangible AR souvenirs that have great potential in creating a personalized and interactive visiting experience.
- We apply different interactive technologies, including AR, hand gestures and voice to make a multi-sensory souvenir. We discuss the design and report on the deployment of the system in the wild.

The hybrid tangible AR souvenir acts as the bridge between the exhibition and the visitors whose memories are tied to the souvenir in a personalized way.

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